

Kakarla Venkata Rama Sai Pavan Shanmukh

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Career Objective

Adaptable B.Tech student and full-stack developer specializing in scalable distributed ledger analytics and secure real-time systems to drive data-driven architectural innovation.

Education

KLEF Deemed to be University 9.51 CGPA	Aug 2023 - Present
Bachelor of Technology (B.Tech) Computer Science and Information Technology	Vijayawada
Narayana Educational Institutions 93.1%	Aug 2021 - Jul 2023
Intermediate Andhra Pradesh Board of Intermediate Education	Vijayawada
Nalanda Vidya Niketan 81.8%	Jul 2020 - May 2021
10th Grade Central Board of Secondary Education	Vijayawada

Experience

MeshaSec Technologies Pvt Ltd.	May 2026 - Present
Cybersecurity Research & AI Intern	
Gained hands-on experience in web application security and backend development by performing vulnerability assessments using DAST tools, solving 10+ security lab machines, and developing/testing REST APIs using Node.js, Express.js, and Postman.	
Cisco Networking Academy	May 2024 - Jul 2024
Virtual Internship	
Hands-on networking with Cisco Packet Tracer. Designed and simulated network topologies; implemented and tested routing algorithms for optimized performance.	

Skills

Core: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Linux, OOPS Through Java, Git and GitHub
Languages: Java, Python, HTML5, Linux, CSS3, JavaScript
Tools, Frameworks and Databases: React.js, Spring Boot, Node.js, Power BI, CI/CD pipelines, MySQL, PostgreSQL

Projects

SentinelStream: Real-Time Ransomware Detection via Behavioral Anomaly Monitoring	Feb 2026 - May 2026
Abstract: A real-time Watchdog-based behavioral monitoring and threshold logic to identify ransomware encryption with 80% accuracy.	
Technologies, Tools & Software: Python, Watchdog, Flask, MySQL, REST APIs, and behavioral anomaly monitoring.	
Outcome and Social Relevance: Minimized detection latency to 10 seconds & safeguarding critical infrastructure against data loss.	
Technologies: Python, React.js, Network Security, JavaScript, Git, GitHub	
OmniBank: Full-Stack Financial Management System with Automated CI/CD	Oct 2025 - Jan 2026
Abstract: A secure full-stack banking platform featuring real-time transactions, MFA, and Automated CI/CD pipelines.	
Technologies, Tools & Software: React.js, Vite, Spring Boot, Spring Security, MySQL, REST APIs, and CI/CD tools.	
Outcome and Social Relevance: Provided a platform for transparent financial tracking.	
Technologies: JavaScript, CI / CD, Docker, Kubernetes, Jenkins	
LexiTree: Optimized Text Analysis via Self-Balancing AVL Trees	Nov 2023 - Dec 2023
Abstract: Decreased AVL Tree's performance to O(log n) word frequency analysis & visualization for large-scale textual datasets.	
Technologies, Tools & Software: AVL Tree, UI Frameworks, and Performance Analysis modules.	
Outcome and Social Relevance: Optimized large-scale data processing.	
Technologies: React.js, Web Technologies, Data Structures and Algorithms, Git, GitHub	

Certifications and Achievements

Microsoft Certified Azure Fundamentals	Mar 2026
Acquired basic knowledge of cloud services and how they are provided via Microsoft Azure.	
Optimized Antenna Design for WBAN Applications	Dec 2024 - Present
A high-efficiency, flexible design featuring a metamaterial-backed ground plane to ensure stable on-body communication.	

Coding Profiles

Leetcode: [klu2300090201](#) Codechef: [kakarla_pavan1](#) Codeforces: [2300090201](#)